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Distance-Vector Routing Algorithm

Choose difficulty:

☒ Beginner

☒ Intermediate

☒ Advanced

What is the primary function of the Distance Vector (DV) routing algorithm?

☐ a: To encrypt network traffic.

☒ b: To compute shortest paths in a network.

☐ c: To assign IP addresses to devices.

☐ d: To manage physical network connections.

Which of the following describes a key characteristic of the Distance Vector algorithm?

☐ a: It requires a central server for all calculations.

☐ b: It is a synchronous and centralized algorithm.

☒ c: It is an iterative and distributed algorithm.

☐ d: It focuses on creating a single, static route.

What information does a node send to its neighbors during the DV update process?

☐ a: Its entire routing table, including host information.

☐ b: Only the cost to its directly connected neighbors.

☒ c: Its own distance vector, which contains cost estimates to all destinations.

☐ d: A list of all other nodes in the network.

What is the purpose of the Bellman-Ford equation in the context of DV routing?

☐ a: To calculate the total number of hops to a destination.

☒ b: To find the cheapest path by considering neighbor costs and their distance vectors.

☐ c: To determine the physical location of a router.

☐ d: To encrypt the distance vector before sending it.

In the Bellman-Ford equation, $D_x(y) = \min_v [c_{x,v} + D_v(y)]$, what does ' $c_{x,v}$ ' represent?

☐ a: The cost from a neighbor 'v' to destination 'y'.

☒ b: The cost to a directly connected neighbor 'v' from node 'x'.

☐ c: The minimum cost from node 'x' to destination 'y'.

☐ d: The cost of the entire path to destination 'y'.

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Distance-Vector Routing Algorithm

Choose difficulty:

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What is the primary goal of the Distance Vector Routing Algorithm?

☒ a: To find the shortest path between two nodes

☐ b: To determine the fastest network protocol

☐ c: To establish direct communication between all nodes

☐ d: To assign IP addresses dynamically

Which type of information does each node maintain in Distance Vector Routing?

☐ a: Only its own IP address

☒ b: Cost estimates to all other nodes

☐ c: A list of all network devices

☐ d: The entire network topology

How do nodes exchange routing information in Distance Vector Routing?

☐ a: By broadcasting their full network topology

☒ b: By exchanging distance vectors with neighbors

☐ c: By sending data packets to all nodes

☐ d: By maintaining a central routing table

Which mathematical equation is used to update a node's distance vector?

☐ a: $D_v(y) = \max(c(x,v), D_x(y))$

☒ b: $D_v(y) = \min(c(x,v), D_x(y))$

☐ c: $D_v(y) = \sum(c(x,v), D_x(y))$

☐ d: $D_v(y) = D_v(y) / c(x,v)$

What happens when a node updates its distance vector?

☐ a: It resets its routing table

☒ b: It immediately sends the updated vector to its neighbors

☐ c: It waits until the next scheduled update

☐ d: It ignores the update unless explicitly requested

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Link state routing

Link State Routing Algorithm

First update:
Link-State - Node U

Node	U	V	W	X	Y	Z
Distance	-	2	5	1	INF	INF

Received Link-State of Node X

Node	U	V	W	X	Y	Z
Distance	1	2	3	-	1	INF

Least-Cost Paths of U

$$D(U) = \min(D(U), [cost(U, V)] + D(V))$$

Node	U	V	W	X	Y	Z
Cost/Path	-	2/U	4/X	1/U	2/Y	INF

Second update:
Least-Cost Paths of U

Node	U	V	W	X	Y	Z
Cost/Path	-	2/U	4/X	1/U	2/X	INF

Link-State - Node U

Node	U	V	W	X	Y	Z
Distance	-	2	5	1	INF	INF

Received Link-state of node Y

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Link State Routing Algorithm

Node	U	V	W	X	Y	Z
Distance	-	2	3	2	INF	INF

Updated Least-Cost Paths of U

$$D(U) = \min(D(U), \{cost(U, V) + D(V)\})$$

Node	U	V	W	X	Y	Z
Cost/Path	-	2/U	3/Y	1/U	2/X	4/Y

NOTE: No change in Costs after receiving this packet from W

Fourth update:

Least-Cost Paths of U

Node	U	V	W	X	Y	Z
Cost/Path	-	2/U	3/Y	1/U	2/X	4/Y

Link-State - Node U

Node	U	V	W	X	Y	Z
Distance	-	2	5	1	INF	INF

Recieved Link-state of node W

Node	U	V	W	X	Y	Z
Distance	5	3	-	3	1	5

Updated Least-Cost Paths of U

$$D(U) = \min(D(U), \{cost(U, V) + D(V)\})$$

Node	U	V	W	X	Y	Z
Cost/Path	-	2/U	3/Y	1/U	2/X	4/Y

NOTE: No change in Costs after receiving this packet from W

Recieved Link-state of node Z

Node	U	V	W	X	Y	Z
Distance	INF	INF	5	INF	2	-

Updated Least-Cost Paths of U

$$D(U) = \min(D(U), \{cost(U, V) + D(V)\})$$

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Link State Routing Algorithm

Shortest DistanceLink StateProtocolPractice

First update:

Link-State - Node U

Node	U	V	X	Y	Z
Distance	-	5	2	INF	INF

Recieved Link-State of Node V

Node	U	V	X	Y	Z
Distance	5	-	INF	1	5

Least-Cost Paths of U

$$D(U) = \min(D(U), \{cost(U, V) + D(V)\})$$

Node	U	V	X	Y	Z
Cost/Path	-	5/U	2/X	6/V	10/Y

Second update:

Least-Cost Paths of U

Node	U	V	X	Y	Z
Cost/Path	-	5/U	2/U	6/V	10/V

Link-State - Node U

Node	U	V	X	Y	Z
Distance	-	5	2	6	10

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Link State Routing Algorithm

Least-Cost Paths of U

Node	U	V	X	Y	Z
Cost/Path	-	5/U	2/U	3/X	10/V

Link-State - Node U

Node	U	V	X	Y	Z
Distance	-	5	2	INF	INF

Recieved Link-state of node Y

Node	U	V	X	Y	Z
Distance	INF	1	1	-	2

Updated Least-Cost Paths of U

$D(U) = \min(D(U), [cost(U, V)] + D(V))$

Node	U	V	X	Y	Z
Cost/Path	-	4/y	2/x	3/y	5/y

Fourth update:

Least-Cost Paths of U

Node	U	V	X	Y	Z
Cost/Path	-	5/U	2/U	3/X	10/V

Link-State - Node U

Node	U	V	X	Y	Z
Distance	-	5	2	INF	INF

Recieved Link-state of node Z

Node	U	V	X	Y	Z
Distance	INF	5	INF	2	-

Updated Least-Cost Paths of U

$D(U) = \min(D(U), [cost(U, V)] + D(V))$

Node	U	V	X	Y	Z
Cost/Path	-	4/y	2/x	3/y	5/y

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Link State Routing Algorithm

Choose difficulty:BeginnerIntermediateAdvanced

How often do nodes broadcast Link-State Packets?

☐ a: Only when the network changes

☒ b: At regular intervals or when the network changes

☐ c: Once every second

☐ d: Only when a data packet is received

Which of the following best describes the forwarding table in Link-State Routing?

☐ a: It stores only directly connected neighbors

☒ b: It maps each destination to the next-hop node on the shortest path

☐ c: It contains the entire network topology

☐ d: It lists all possible paths to a destination

What is the primary advantage of Link-State Routing over Distance Vector Routing?

☐ a: Less memory usage

☒ b: Faster convergence and no count-to-infinity problem

☐ c: Lower processing power required

☐ d: Better suited for small networks only

How does a node update D(v) for a neighbor v in the Link-State algorithm?

☐ a: $D(v) = \max(D(v), D(w) - c(w,v))$

☐ b: $D(v) = D(w) / c(w,v)$

☒ c: $D(v) = \min(D(v), D(w) + c(w,v))$

☐ d: $D(v) = \text{sum}(D(v), c(w,v))$

What is the final output of the Link-State algorithm?

☒ a: A complete routing table containing next-hop nodes

☐ b: A set of shortest paths but no forwarding information

☐ c: A list of all network devices

☐ d: A distance vector for each node

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